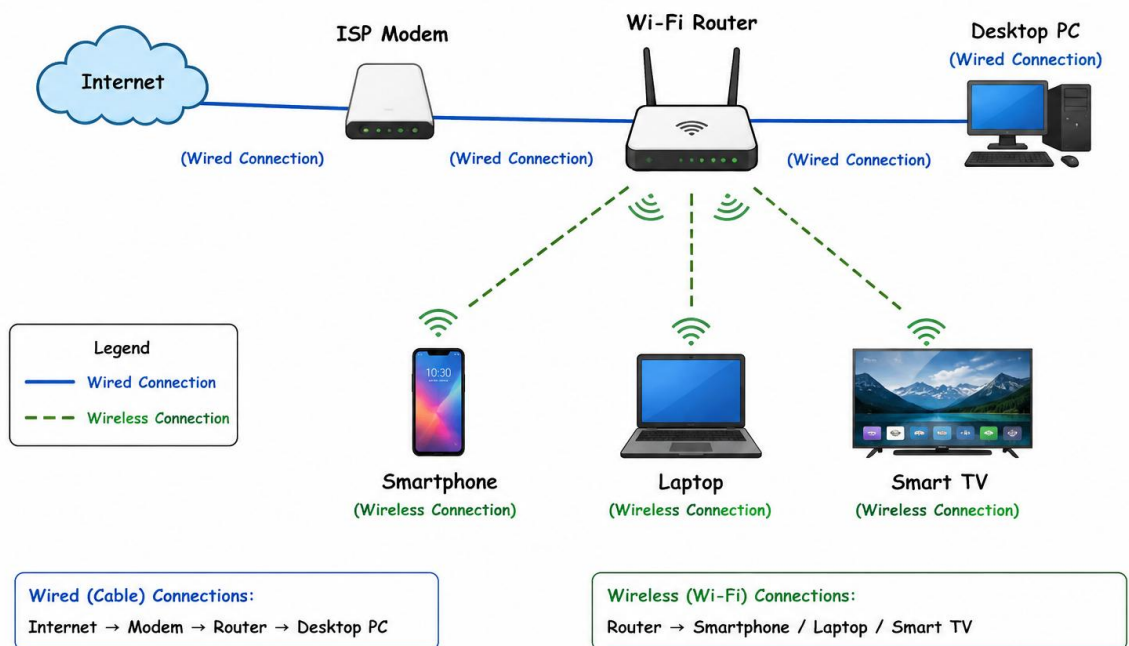


Task 1: Popular Wi-Fi Standards

Wi-Fi Standard	Frequency Band	Maximum Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz & 5 GHz	Up to 600 Mbps	Supports both 2.4 GHz and 5 GHz, offering better range than older standards.
802.11ac (Wi-Fi 5)	5 GHz	Up to 3.5 Gbps	Faster than 802.11n and uses only the 5 GHz band for higher performance.
802.11ax (Wi-Fi 6)	2.4 GHz & 5 GHz (and 6 GHz with Wi-Fi 6E)	Up to 9.6 Gbps	Provides higher speed, lower latency, and better performance with many connected devices.

Task 2: Home Wi-Fi Network Diagram

Smartphone connects to Internet via Wi-Fi Router at Home



Task 3: Wi-Fi Scan Results (Example)

SSID	Signal Strength	Security Type	Recommendation
Home_WiFi	-42 dBm (Excellent)	WPA3	Best Choice
Neighbor_5G	-65 dBm (Good)	WPA2	Good, but not preferred
CoffeeShop	-80 dBm (Weak)	Open (No Password)	Not Recommended

Which network would I choose?

I would choose Home_WiFi because:

- It has the strongest signal (-42 dBm), providing faster and more stable connectivity.
- It uses WPA3 security, which offers better protection than WPA2.
- It is a private and trusted network, making it safer than an open public Wi-Fi network.

Task 4: SSID Hiding

SSID hiding is a feature that prevents a Wi-Fi router from broadcasting its network name (SSID). As a result, the network does not appear in the list of available Wi-Fi networks. Users must manually enter the network name and password to connect.

Advantage

- Adds a small layer of privacy by making the network less visible to casual users.

Disadvantage

- It is not a strong security measure, because the hidden SSID can still be detected using network scanning tools. It can also make connecting new devices less convenient since users must manually enter the network name and password.